

HUYNH TRUNG NGUYEN

AI ENGINEER

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ABOUT ME

AI Engineer specialized in building production-grade LLM-powered applications and multi-agent orchestration systems using LangGraph and FastAPI. Experienced in implementing complex agentic workflows, RAG systems, real-time streaming architectures, and performance optimization for inference pipelines.

WORK EXPERIENCE

AI Automation

04/2025 - 06/2025

Tin Hoc Sao Viet Company

- Designed and deployed automated workflows using n8n, integrating multiple APIs and services to streamline business processes while leveraging LLM capabilities for intelligent document processing.
- Debugged and optimized workflow performance, identifying bottlenecks and implementing solutions to ensure reliable automation execution and improved operational efficiency.

SKILLS

- Programming Languages:** Python, JavaScript.
- AI & Machine Learning:** PyTorch, LangChain, LangGraph, LangSmith, Scikit-learn, NumPy, Pandas.
- Backend & API Development:** FastAPI, n8n.
- Data Visualization & Analysis:** Matplotlib, Plotly.
- Databases:** MySQL, PostgreSQL, MongoDB.
- Version Control:** Git, GitHub.

FEATURED PROJECTS

Multi Agent Writer

08/2026 - 01/09/2026

[GitHub](#) ↗ | [Live Demo](#) ↗

Technologies: Python, LangGraph, LangChain, LangSmith, FastAPI, PostgreSQL, MCP, Tavily, Wikimedia Commons, Gemini, Groq

- Multi-Agent LangGraph Workflow:** Architected an 8-node LangGraph pipeline (Guardrail → Supervisor → Planner → HITL → Executor → Image Resolver → Synthesizer → Evaluator) with clear separation of concerns, enabling specialized agents to handle distinct responsibilities in the content generation process.
- Intelligent Task Dependency DAG:** Implemented task dependency graph execution using Kahn's algorithm, allowing parallel task processing while respecting dependency constraints. Tasks with no dependencies execute simultaneously, maximizing throughput and reducing total workflow duration.

- **Native Human-in-the-Loop (HITL):** Integrated LangGraph's native interrupt() mechanism combined with Command(resume=...) to enable users to approve, edit, or reject generated plans, with automatic approval fallback after 5 minutes.
 - **FastAPI Backend Orchestration:** Built backend services to execute LangGraph workflows asynchronously, handle workflow interrupts/resumes, manage PostgreSQL checkpoint storage for state persistence, and provide real-time progress tracking.
 - **MCP & Caching:** Integrated MCP tools for Tavily research and Wikimedia image search, with PostgreSQL caching using TTL policies to reduce redundant API calls and optimize workflow performance.
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Sunny Agent (Agentic Chatbot)

07/2026 - 08/2026

[GitHub ↗](#) | [Live Demo ↗](#)

Technologies: Python, LangGraph, LangChain, LangSmith, FastAPI, Chroma, SQLite, Tavily, SSE, Gemini, Groq

- **LangGraph-Based Conversational Agent:** Designed a LangGraph workflow (Input Guardrail → Chatbot → Tools → Review → Output Guardrail) orchestrating multi-tool agent interactions.
- **Real-Time Streaming Backend:** Implemented FastAPI endpoints with Server-Sent Events (SSE) for progressive response streaming, combined with SQLite-based conversation history persistence.
- **Document-Based RAG:** Built a document ingestion pipeline with Chroma vector database for semantic search and integrated RAG retrieval into the agent's tool-calling workflow.
- **Safety & Quality Guardrails:** Implemented input/output guardrails and an automatic response review and rewriting mechanism to improve response safety and consistency.

CERTIFICATIONS

English CEF B1

06/2025

- Ho Chi Minh City University of Technology

EDUCATION

Bachelor of Science in Information Technology

09/2021 - 03/2026

- Ho Chi Minh City University of Technology